GEOPHYSICS

Palæomagnetism of Some Dolerite Intrusions from the Theron Mountains and Whichaway Nunataks, Antarctica

As a part of the scientific activity of the Commonwealth Trans-Antarctic Expedition a collection of eight oriented samples was made by Stephenson from dolerite sills and dykes in the region of the Theron Mountains, Shackleton Range and Whichaway Nunataks. The seven intrusions which have been sampled bear petrological affinities with the Karroo Dolerites of South Africa and with the dolerite sills of Tasmania, and are thought to be of comparable age. They intrude flat-lying sediments and show no evidence of post-formational tilting.

The directions of remanent magnetization have been measured by Blundell and are shown in Fig. 1. These are arranged in two groups, normal and reversed, which are 180° apart. The two groups have been combined by reversing the polarity of those with downward dips to provide a mean direction of magnetization with azimuth 64° E. of N. and dip 68° up. The circle of 95 per cent confidence, applying Fisher's statistics[1], has a radius of 12 deg.

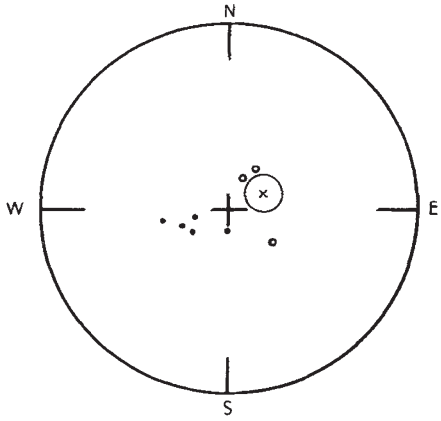


Fig. 1. Stereogram of directions of magnetization of Antarctic Dolerite intrusions. ○, dip up ; ●, dip down ; ×, mean direction with circle of 95 per cent confidence

Storage tests similar to those already described[2] were applied during the measurements of magnetic direction and indicated a stable magnetization. It is also evident from Fig. 1 that there is a high degree of stability, for the normal and reversed groups have remained 180° apart and at a substantial angle to the present field direction. The samples are unaltered chemically and there is no evidence of the development of any secondary magnetization, so it may be presumed that the mean direction of magnetization is a true record of the field direction when the rocks were intruded. Since seven intrusions from a wide area were sampled, the effects of local variations in the geomagnetic field and of secular variation are unlikely to be significant.

Over a period of 1,000 years or more, the geomagnetic field appears on average to resemble that of a geocentric axial dipole. On this basis the position of the (south) pole relative to the Theron Mountains when the sills were intruded was at lat. 54° S., long. 136° W. with an oval of 95 per cent confidence of 16° radius. This pole position may be compared with those similarly derived for the Karroo Dolerites

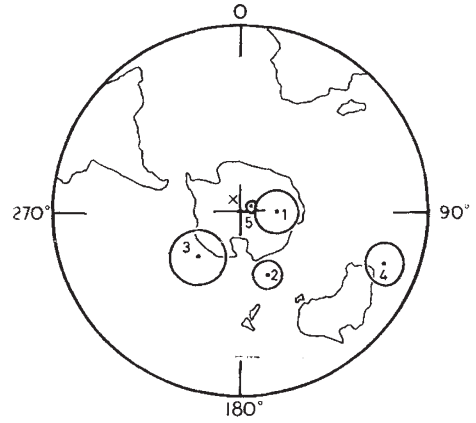


Fig. 2. Jurassic Poles for Africa (1), Australia (2), Antarctica (3), India (4) and South America (5), with ovals of 95 per cent confidence. ×, sampling area

of South Africa[3], the Tasmanian Dolerites[4], the Rahajmahal Traps of India[5] and the Serra Geral lavas of Brazil[6], shown in Fig. 2. The age correlation between these rocks is not good, but it may be stated with some confidence that they are all Jurassic, and are the same age to within 30 million years. The difference between the Antarctic and Indian pole positions is more than 100°, and though this could be explained in terms of Gold's mechanism[7] for polar wandering, it seems unlikely in view of the relatively small variation in pole positions from the Jurassic to the present day relative to a single continent. The scatter in pole positions shown in Fig. 2 is more indicative of relative movement of the continents, and it is interesting to note that these results are not inconsistent with earlier reconstructions put forward on independent evidence by proponents of the continental drift hypothesis.

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Palæomagnetism in Some Norwegian Sparagmites and the Late Pre-Cambrian Ice Age

Pre-Pleistocene glacial deposits have been claimed from many ages and latitudes. Palæomagnetic studies of the Permian tillites in Australia[1] suggest that they were near the geomagnetic, and probably also the geographical pole. The late Pre-Cambrian ('Eo-Cambrian', Varangian, Sinian, etc.) tillites are of special interest for several reasons :

(1) Widespread occurrences have been claimed[2] and many are well established, for example, in Australia, China, Greenland, Norway and Spitsbergen.

(2) Their stratigraphical position suggests that all are referable to one major glacial period (with, however, interglacial or interstadial intervals). It is therefore a key horizon in international correlation below the fossiliferous Lower Cambrian.

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(3) The boulder beds usually suggest deposition from floating ice in an apparently continuous marine succession.

(4) Palæomagnetic determinations of Palæozoic and late Pre-Cambrian rocks from Britain and America[3] give pole positions which suggest that the Norwegian and Arctic tillites were deposited not far from the geomagnetic equator. The successions in which they occur show facies consistent with a near equatorial position.

In the course of investigations of the magnetism of some Arctic rocks, collections of tillites and associated horizons were made by us on Dr. Lauge Koch's East Greenland Expedition in 1957, and from Spitsbergen and southern Norway in 1958. This communication gives the results of the smallest of these (29 specimens collected from the sparagmites on Lake Mjösen in southern Norway by W.B.H.) and discusses them in the light of work on the Greenland and Spitsbergen material.

Holtedahl first suggested a glacial origin for some of the Moelven conglomerates[4] and the succession sampled was first mapped by Goldschmidt[5]. Table 1 gives the succession as summarized by Holtedahl[6] and the stratigraphical position of the samples collected. We are indebted to Dr. T. Strand of Oslo for advice as to the best localities. The rocks were generally coarse, and would not have been selected for palæomagnetic work but for their critical stratigraphical position. Samples were orientated in the field by rock compass[7]. In the laboratory the rocks were re-oriented, cored, and measured on the astatic magnetometer, similar in construction to that described by Collinson *et al.*[8], which is housed at the Department of Geodesy and Geophysics, Cambridge. For these facilities we wish to thank Mr. B. C. Browne and Dr. J. C. Belshé. From each rock three samples were measured, the vectors combined by meaning the direction cosines and a statistical estimate of the dispersion made using the method proposed by Fisher[9]. These mean vectors were then combined to give the pole position[10], the main computations being carried out on the Cambridge digital computor *Edsac*.

TABLE 1

	Horizon	No. of sites	No. of specimens	No. of specimens giving significant results
9	Ringsaker Quartzite	3	9	3
8	Vardal Sparagmite	2	5	3
7	Ekre Shale			
6	Moelv Conglomerate and Tillite	3	6	4
5	Moelv Red Sparagmites	1	1	1
4	Biri Limestone			
3	Biri Conglomerates	2	5	—
2	Bröttum Limestones and Shales			
1	Bröttum Grey Sparagmites	1	1	—

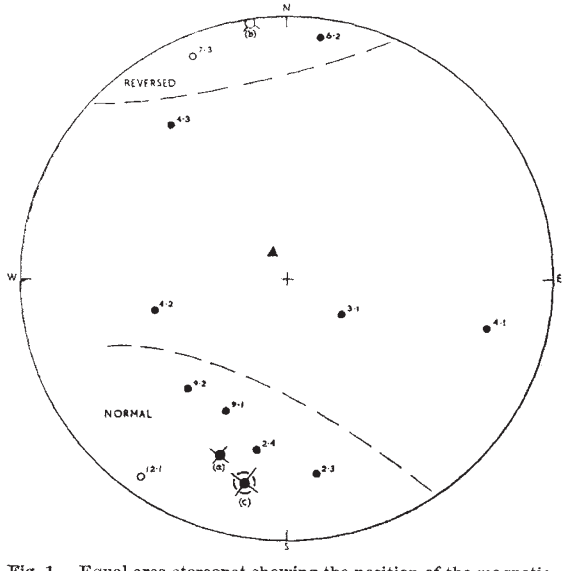


Fig. 1. Equal area stereonet showing the position of the magnetic vectors for eleven rocks from the Norwegian tillites together with the mean vectors derived from the normal, reversed, and the combined normal and reversed groups. ●, Vector plotting in lower hemisphere ; ○, vector plotting in upper hemisphere ; *a*, mean vector normal group, lower hemisphere ; *b*, mean vector reversed group, upper hemisphere ; *c*, mean vector of combined normal and reversed groups in lower hemisphere ; ▲, direction of present magnetic field

TABLE 2

Specimen No.	Horizon No.	Locality	Field Position D	Field Position I	Dip Corrected D	Dip Corrected I	α	κ	Intensity (10^{-6} G./c.c.)
MN 07·3	9	Coast N. of Evjeviken	334	29	377	−8	20	38	1·4
MN 02·3	9	Coast Skotta	172	−28	171	24	14	83	13·7
MN 02·4	9	Coast Skotta	186	−8	190	33	32	16	6·2
MN 06·2	8	N.E. of Moelven	042	72	008	7	4	896	3·8
MN 09·1	8	Coast S. of Smedstad-viken	198	10	205	42	16	57	1·6
MN 09·2	8	,, ,,	211	13	221	43	17	51	0·4
MN 03·1	6	Coast N. of Tyvehl	151	27	124	70	18	49	1·1
MN 04·1	6	Coast S. of Tyvehl	120	20	105	23	12	109	3·3
MN 04·2	6	,, ,,	237	23	257	47	4	993	1·0
MN 04·3	6	,, ,,	275	53	324	28	11	122	10·3
MN 12·1	1	E. of Bergsodden	217	−7	215	−6	9	194	12·1

The following abbreviations are used : *D* declination in degrees measured from true north ; *I* inclination to the horizontal in degrees, positive downwards, negative upwards ; α semi-angle of the cone of confidence drawn at 95% probability level ; κ estimate of precision.

Of the twenty-nine rocks collected, eighteen were too weakly magnetized and too inhomogeneous to give significant results. The remaining eleven specimens gave the mean vectors listed in Table 2 and plotted on the equal area stereonet in Fig. 1 after correction for observed geological dip. The vectors fall into two groups.

The first group consists of four rocks collected at two sites with mean vectors which show a considerable scatter but with a mean direction :

D 352°, *I* 87°, α89°, κ2

which lies very close to the direction of the present magnetic field in the area. We think these rocks have been partially magnetized in the direction of the present field.

The second group consists of five vectors lying in the south-west quadrant and two vectors which are nearly opposite in position, but which if reversed would fall close to the first five vectors. We think that this was caused by the Earth's magnetic field reversing its polarity at least twice during the time span covered by the rocks collected. Combining the five normal vectors with the two reversed vectors, after these have been re-reversed, gives a mean direction :

D 191°, *I* 21°, α24°, κ7.

This gives a position for the corresponding magnetic pole at :

Lat. 18°N., Long. 179°E.

Despite the relative uncertainty due to the small number of rocks and their generally weak magnetization, the resulting pole position is close to that for the Cambrian of Great Britain[11] and also to tillites from Greenland collected by us (Fig. 2), and intermediate to the pole positions for rocks of Silurian and Late Pre-Cambrian ages from the United States[3]. In all these cases, and also for the tillites of comparable age collected from Spitsbergen and at present

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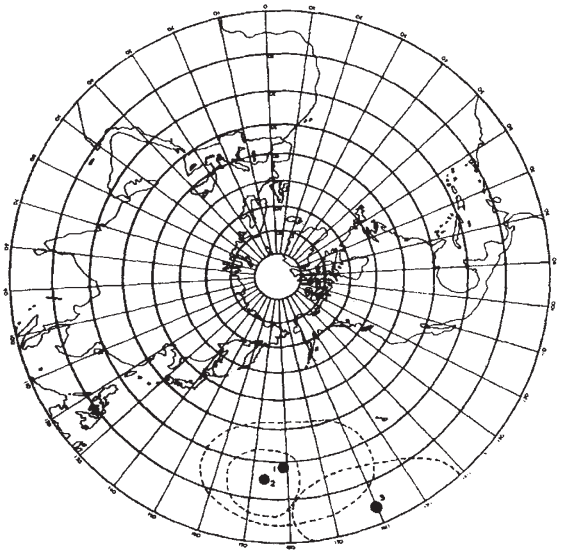


Fig. 2. Polar projection showing positions of magnetic poles derived from 1, Norwegian tillites; 2, Caerbwdy Sandstones Wales; 3, Greenland tillites. The broken lines show the ovals of confidence drawn at the 95 per cent level.

being investigated, the collecting areas appear to have lain in equatorial regions not far removed from the earth's geomagnetic equator. If it be assumed that the Earth's magnetic pole, when averaged over long periods, has been close to the geographical pole, the palæomagnetic evidence suggests that at the close of Pre-Cambrian time Scandinavia, Spitsbergen and Greenland were equatorial rather than polar regions.

Pole positions for the tillite horizons are therefore consistent with the palæomagnetic picture already obtained, at least as regards latitude, and it is probable that marine tillites were formed near the equator in late Pre-Cambrian time. This evidence, together with the world-wide distribution of supposed tillites of the same age, argues an ice age far more severe than any later ice age. The thickness of many tillite deposits, together with some evidence from Greenland rocks of appreciable change in the geomagnetic field during the tillite period, suggest a duration many times that of the Quaternary Ice Age to date.

Such an event may have profound significance in the understanding of late Pre-Cambrian–Cambrian stratigraphy. In particular, the sudden widespread appearance of highly organized life in Lower Cambrian times may in part be the result of a general relaxation of prolonged severe glacial control both of faunas and sea level as already suggested[12]. The physiographical and astronomical implications of such a glaciation are also worthy of note.

The extension downwards of an internationally acceptable stratigraphical system beneath the Cambrian is now much discussed because of the similarity of many transitional successions throughout the world. For this the glacial horizons, already recognized as providing a stratigraphical datum, may yield a succession capable of correlation and worthy of status as an infra-Cambrian system (?Varegian[13]). The establishment of new systems is premature without adequate correlation and a recent conference[14] concluded that radioactive dating might provide such a basis. Our work suggests that palæoclimatic and palæomagnetic correlation together promise a more immediate means of correlation within these late Pre-Cambrian successions.

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METALLURGY

X-Ray Diffraction Pattern of Carbide in Low-Carbon Iron-Silicon Alloys

A CARBIDE which has a needle-like appearance on a polished and etched cross-section is a common occurrence in hot-rolled 3·25 per cent silicon iron. It is metastable with respect to cementite and can be formed by a variety of heat treatments[1].

An electron-transmission diffraction pattern was obtained from a disk (carbide single crystal) separated from a commercial 3·25 per cent silicon steel containing 0·03 per cent carbon[1]. This pattern could be indexed on the basis of a cubic or tetragonal structure with $a_0 = 6{\cdot}06$ A. It was concluded, however, that these structures were unlikely because of a large number of omissions in the pattern, and that the crystal structure might be complex with a large unit cell. It was obvious that the carbide was neither cementite nor epsilon carbide ($Fe_{2\cdot4}C$).

A number of attempts were made to obtain a reproducible X-ray diffraction pattern from hot-rolled 3·25 per cent silicon steel, containing 0·03 per cent carbon or less. All these failed, presumably because the amount of carbide present was insufficient. To obtain a reproducible pattern, it was necessary to increase the carbon to 0·06 per cent, dissolve it in ferrite at a high temperature, quench, then age to develop the precipitate.

A 2-lb. vacuum melt containing 0·062 per cent carbon and 3·51 per cent silicon was prepared. To dissolve all the carbide in ferrite, samples of this alloy were heated to 2,540°F. (1,395°C.) for ½ hr. in evacuated silica capsules. After brine quenching, the samples were aged at 300°F. (150°C.), 600°F. (315°C.) or 800°F. (425°C.) for various periods. These treatments produced a large quantity of oriented precipitate (Fig. 1). After ageing, blanks from each sample were machined to small cylinders, 1 mm. diameter × 5 mm. long, then etched in nitric acid to 0·5 mm. diameter. Adjacent pieces were prepared for metallographic examination. The specimens for X-ray examination were given a final etch in picral, nital or 2 per cent iodine ethyl alcohol to put the precipitate at the surface in relief, then washed in methanol and dried. These specimens were mounted in a 114 mm. Debye-Scherrer camera and exposed to iron-filtered cobalt radiation at 40 kV. and 10 m.amp. for various periods. Since several grains in each sample were irradiated, a

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